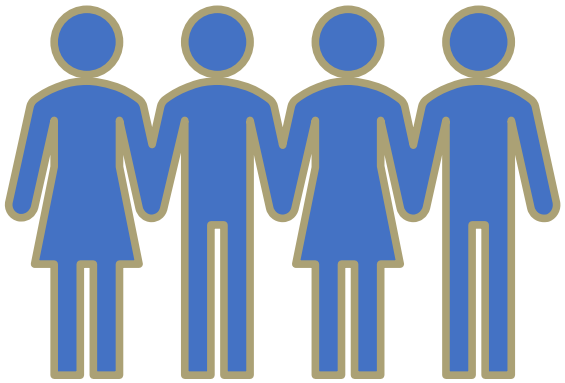
Constrained Growth and Decay



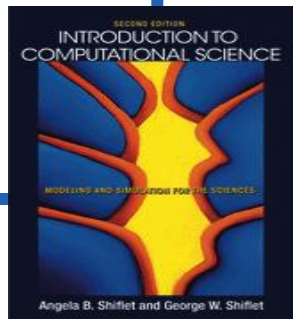
Module 2.3
Constrained Growth

Angela B. Shiflet and George W. Shiflet

Wofford College

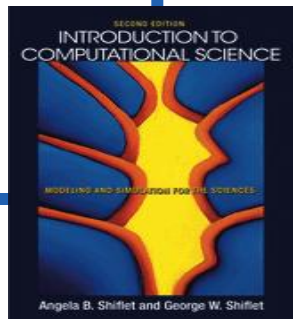
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Unconstrained Growth and Decay of population (P)

- $dP/dt = rP$
- Limitations to unconstrained growth?
- Carrying capacity (M): maximum number of organisms area can support

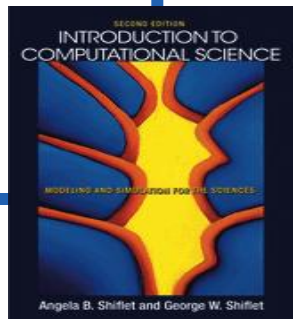


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Unconstrained Growth and Decay of population (P)

- Populations increase rapidly (such as exponential growth) at first, but they eventually encounter **resistance** from the environment – competitors, predators, limited resources, and disease.
- The maximum population size that the environment can support indefinitely is called the **carrying capacity**.

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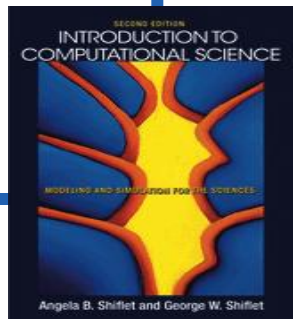
Rate of change of population

- D = number of deaths
- B = number of births
- rate of change of P = (rate of change of B) – (rate of change of D)

$$\frac{dP}{dt} = \frac{dB}{dt} - \frac{dD}{dt}$$



- Rate of change of population:
- $\frac{dP}{dt} = rP - \frac{dD}{dt}$



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Constrained Growth

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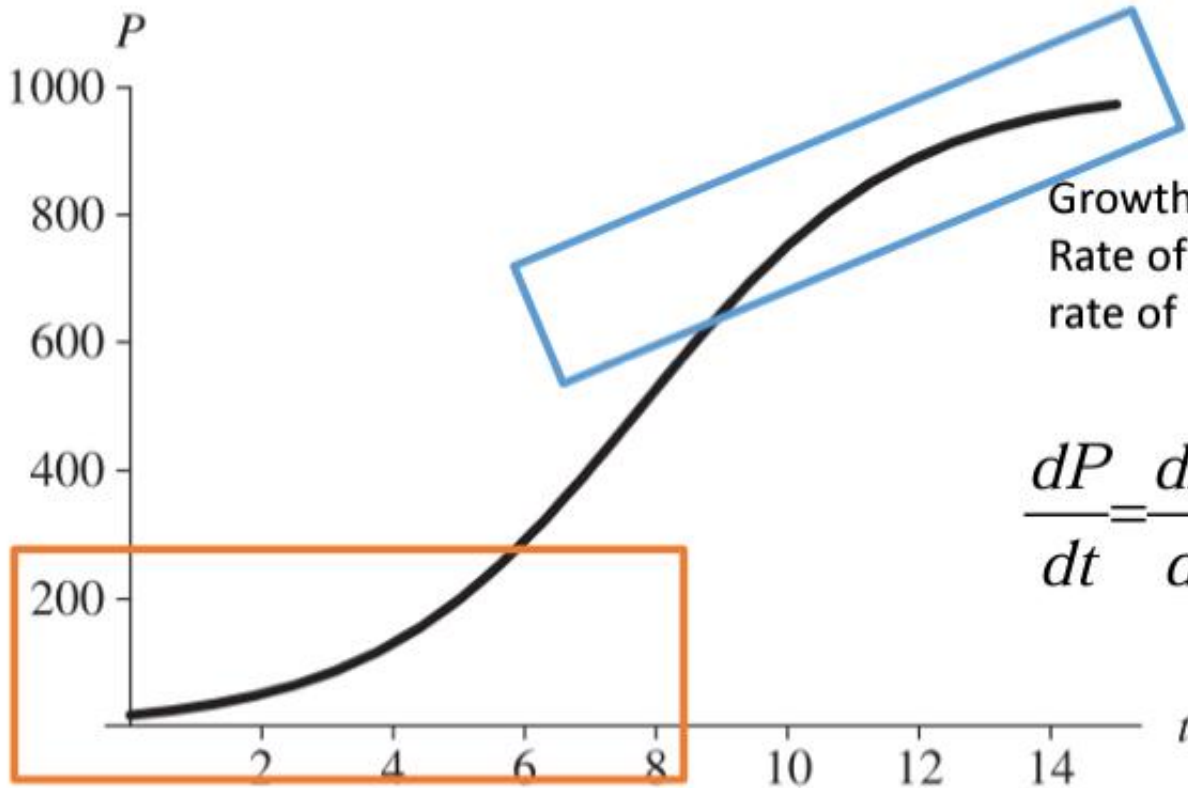
If population is less th
but close to carrying c

$$\frac{dP}{dt} \approx \frac{dB}{dt} = rP$$

If population is much less than carrying capacity

Almost unconstrained model

Rate of change of D $(\frac{dD}{dt}) \approx 0$



Growth is dampen/lessen, aln
Rate of change of D almost eq
rate of change B

$$\frac{dP}{dt} = \frac{dB}{dt} - \frac{dD}{dt} \approx rP - rI$$

$$\frac{dD}{dt} \approx \frac{dB}{dt}$$

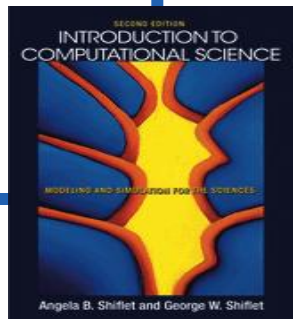
If population is much less than carrying capacity

- Almost unconstrained model
- Rate of change of D (dD/dt) ≈ 0

$$\frac{dD}{dt} \gg 0$$

$$\frac{dP}{dt} \gg \frac{dB}{dt}$$

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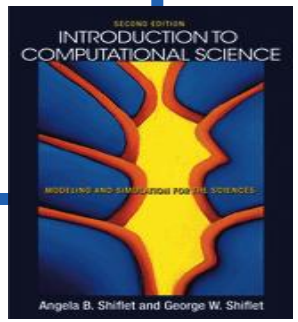


If population is less than but close to carrying capacity

- Rate of change of D larger, approaching rate of change B
- Growth is dampened, almost 0

$$\frac{dD}{dt} \gg \frac{dB}{dt} = rP$$

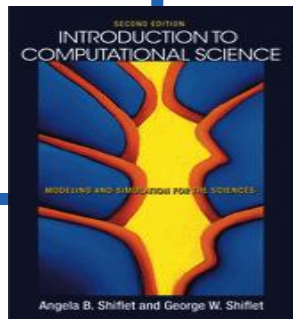
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$$\frac{dP}{dt} = rP - \frac{dD}{dt}$$

$dD/dt = death_rateP$,
multiply $death_rateP$ by something so that

- $dD/dt \approx 0$ for P much less than M
  - In this situation, f almost 0
- $dD/dt \approx dB/dt = rP$ for P less than but close to M
  - In this situation, f almost1
- What is a possible factor f?
  - One possibility is P/M



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Rate of change of population

- Rate of change of B proportional to P

$$\frac{dB}{dt} = rP$$

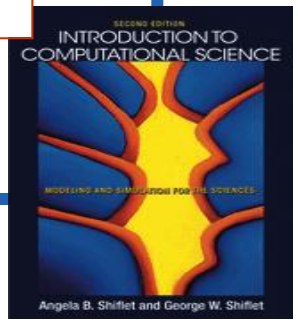
- Rate of change of D, by including carrying capacity M

$$\frac{dD}{dt} = \left(r\frac{P}{M}\right)P$$

$$\frac{dP}{dt} = \frac{dB}{dt} - \frac{dD}{dt}$$

$$\frac{dP}{dt} = \underbrace{(rP)}_{births} - \underbrace{\left(r\frac{P}{M}\right)P}_{deaths}$$

$$\frac{dP}{dt} = r\left(1 - \frac{P}{M}\right)P$$



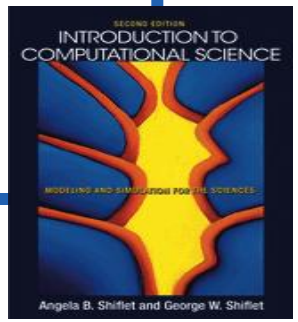
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Continuous logistic equations

$$\frac{dD}{dt} = (rP)\frac{P}{M}$$

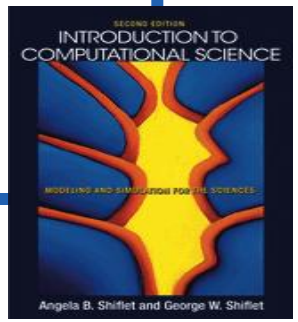
$$\frac{dP}{dt} = (rP) - (rP)\frac{P}{M} = r\left(1 - \frac{P}{M}\right)P$$

This **logistic equation** can be solved by using separation of variables.



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- The logistic equation (sometimes called the Verhulst model or logistic growth curve) is a model of population growth first published by Pierre Verhulst (1845, 1847).

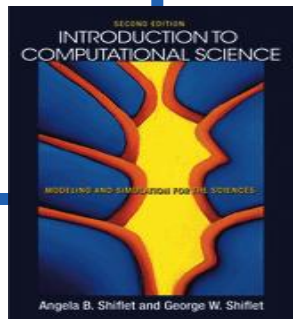


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Continuous logistic equations

If we let a = r and b = r/M, then the logistic equation can be rewritten as dP/dt = P(a – bP). Use separation of variables and partial fraction decomposition, we have $(\frac{\frac{1}{a}}{P} + \frac{\frac{b}{a}}{a-bP})\mathrm{dP} = \mathrm{dt}$, integrate and simplify, we have $P(t) = \frac{aP_0}{bP_0+(a-bP_0)e^{-at}}$ where $P_0 = P(0)$.

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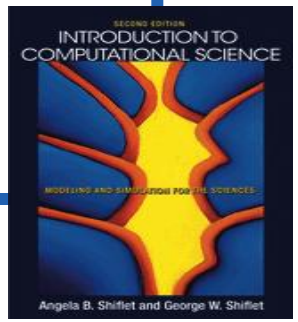
Discrete logistic equations

$$\Delta D = D(t) - D(t - \Delta t) = \left(rP(t - \Delta t)\right)\frac{P(t - \Delta t)}{M}\Delta t$$

$$\Delta P = \text{births} - \text{deaths}$$

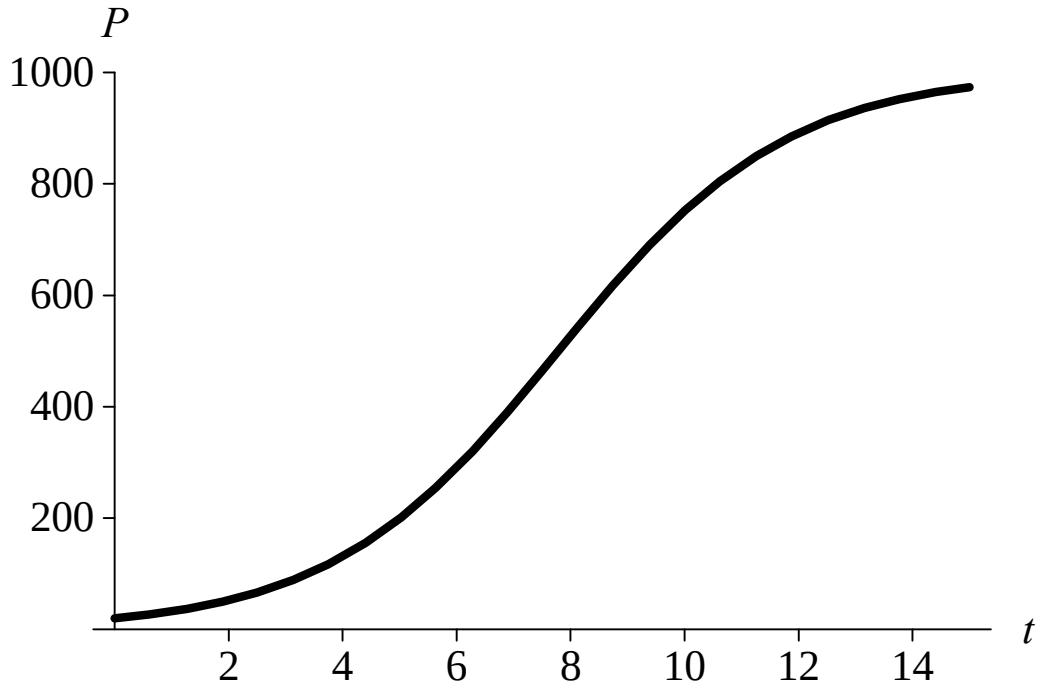
$$\Delta P = \left(rP(t - \Delta t)\right)\Delta t - \left(rP(t - \Delta t)\right)\frac{P(t - \Delta t)}{M}\Delta t$$

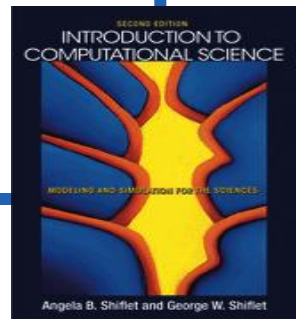
$$\Delta P = k\left(1 - \frac{P(t - \Delta t)}{M}\right)P(t - \Delta t), \text{ where } k = r\Delta t$$



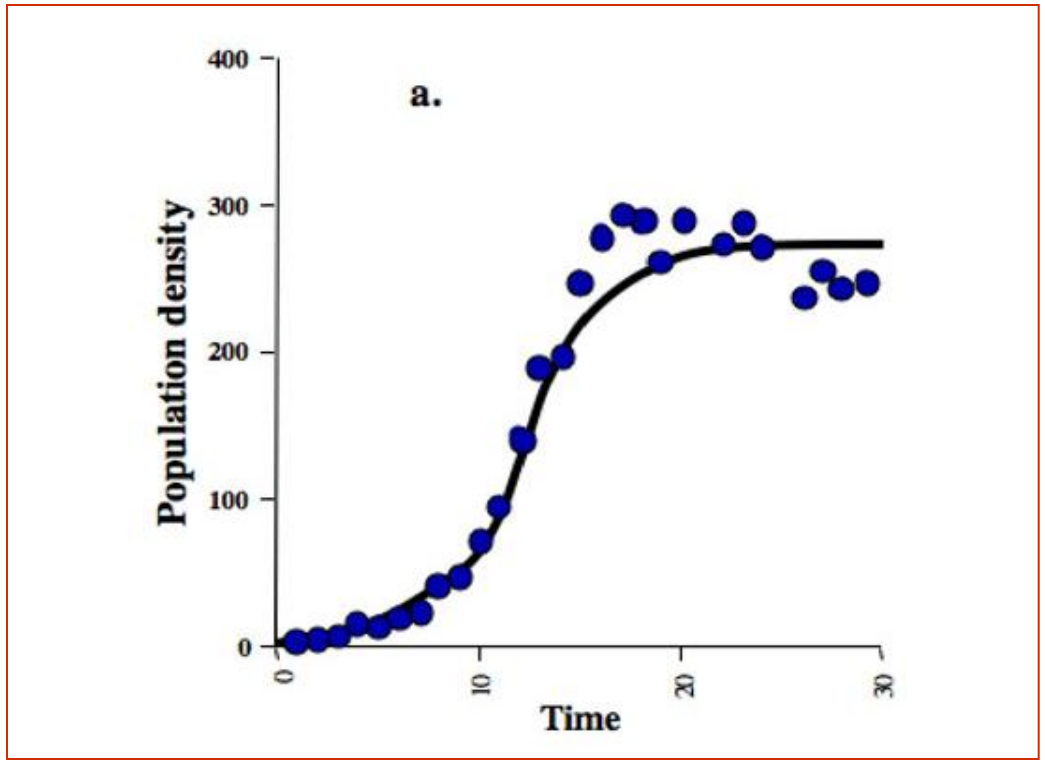
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If initial population < M, S-shaped graph





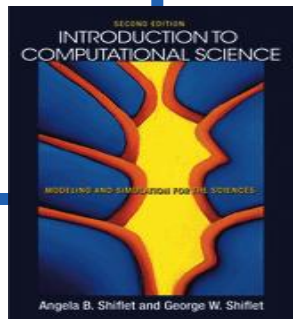
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A laboratory population of Paramecium

Note how the density first looks exponential (indeed these are the same data presented in figure 2, but over a longer time frame), but later, after the population gets to around 300 cells per cc, it levels off. This is the classic logistic pattern.

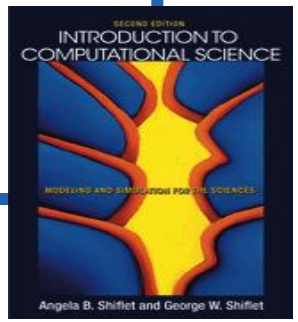
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If population greater than *M*

- What is the sign of growth?
  - Negative (P decreases in size through deaths)
- How does the rate of change of *D* compare to the rate of change of *B*?
  - Greater
- Does this situation fit the model?

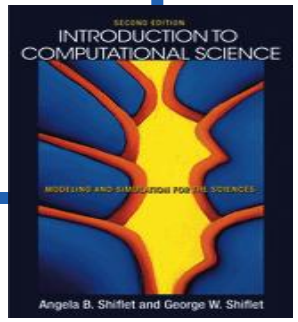
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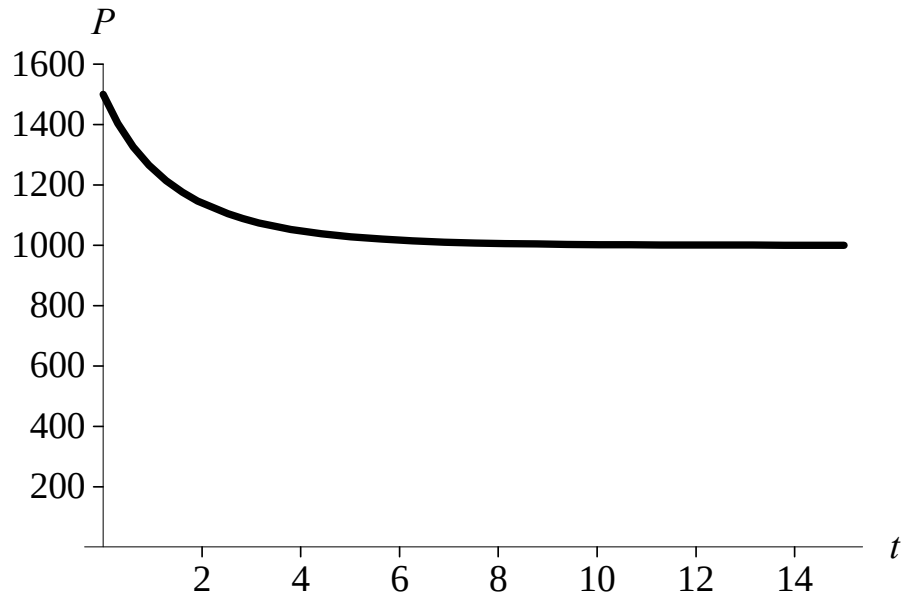
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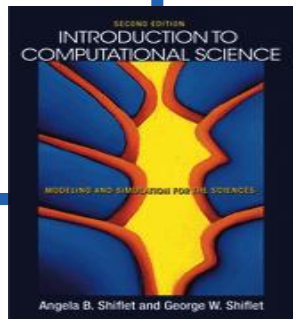
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If initial population > M



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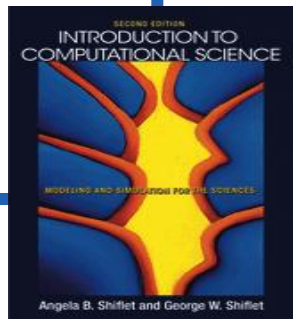


The logistic equation with carrying capacity $M = 1000$ has an interesting property.

- If the initial population is less than 1000, as in Figure 2.3.1, the population increases to a limit of 1000.
- If the initial population is greater than 1000, as in Figure 2.3.2, the population decreases to the limit of 1000.
- Moreover, if the initial population is 1000, we see that

$$\frac{P}{M} = \frac{1000}{1000} = 1 \text{ and } \frac{dP}{dt} = r(1-1)P = 0.$$

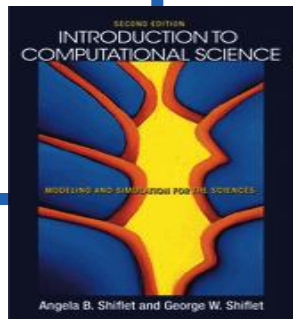
- In discrete terms, $\Delta P = 0$.
- A population **starting at the carrying capacity remains there**.
- We say that $M = 1000$ is an **equilibrium** size for the population because the population remains steady at that value or $P(t) = P(t - \Delta t) = 1000$ for all $t > 0$.



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Definitions An **equilibrium solution** for a differential equation is a solution where the derivative is always zero. An **equilibrium solution** for a difference equation is a solution where the change is always zero.

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Stability

- Equilibrium solution q is stable if there is interval (a, b) containing q, such that if initial population $P(0)$ is in that interval then
  - $P(t)$ is finite for all $t > 0$
  - $P \rightarrow q$
- $P = M$ is stable equilibrium
- $P = 0$ is unstable equilibrium

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